

1 Norm Approximation

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$. The objective is to minimize the residual vector $\mathbf{r} = \mathbf{Ax} - \mathbf{b}$.
The general problem is formulated as,

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{Ax} - \mathbf{b}\|.$$

Depending on the norm chosen, this results in different convex optimization problems with unique statistical interpretations and solution characteristics.

1.0.1 ℓ_1 Norm Approximation (Sum of Absolute Residuals)

$$\|Ax - b\|_1 = \sum_{i=1}^m |(Ax - b)_i|$$

Also known as **Least Absolute Deviations (LAD)**.

- **Robustness:** Unlike the ℓ_2 norm, the ℓ_1 norm places less weight on large residuals (outliers). It yields a robust estimator where the solution tends to minimize the median of the residuals rather than the mean.
- **Sparsity:** In problems like Basis Pursuit (where we minimize $\|x\|_1$ subject to constraints), ℓ_1 induces sparsity. Here, in residual minimization, it allows many residual entries to be exactly zero.

Convex Formulation (Linear Program): The problem is non-differentiable at zero. We reformulate it as a Linear Program (LP) by introducing slack variables $t \in \mathbb{R}^m$:

$$\begin{aligned} & \text{minimize} && \mathbf{1}^T t \\ & \text{subject to} && -t \preceq Ax - b \preceq t \end{aligned}$$

where $\preceq$ denotes component-wise inequality.

1.0.2 ℓ_2 Norm Approximation (Least Squares)

$$\|Ax - b\|_2 = \left(\sum_{i=1}^m (Ax - b)_i^2 \right)^{1/2}$$

Minimizing the ℓ_2 norm is equivalent to minimizing the squared Euclidean norm $\|Ax - b\|_2^2$. This is the standard **Least Squares** problem.

- **Statistical Interpretation:** Corresponds to the Maximum Likelihood Estimator (MLE) under the assumption of Gaussian noise.
- **Sensitivity:** Highly sensitive to outliers because errors are squared.

Convex Formulation (Unconstrained Quadratic): This is a differentiable, unconstrained convex problem with a closed-form analytical solution (assuming A is full rank):

$$x^* = (A^T A)^{-1} A^T b$$

1.0.3 ℓ_∞ Norm Approximation (Chebyshev)

$$\|Ax - b\|_\infty = \max_{1 \leq i \leq m} |(Ax - b)_i|$$

Also known as **Minimax** or **Chebyshev approximation**.

- **Goal:** Minimizes the worst-case residual.
- **Applications:** Critical in control design and engineering where ensuring that error never exceeds a safety threshold is more important than the average error.

Convex Formulation (Linear Program): We introduce a scalar variable $t \in \mathbb{R}$ representing the maximum bound:

$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && -t\mathbf{1} \preceq Ax - b \preceq t\mathbf{1} \end{aligned}$$

1.1 Penalty Functions

In many approximation problems, we minimize a sum of penalty functions applied to the residuals $r_i = (Ax - b)_i$. The problem takes the form:

$$\min_x \sum_{i=1}^m \phi(r_i)$$

where $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex penalty function.

1.1.1 Norm Based ($p = 1, 2, \infty$)

$$\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty.$$

1.1.2 Dead Zone Linear

Functions of the form, for $r \in \mathbb{R}^n$,

$$\sum_i \phi(r_i),$$

where

$$\phi(r) = \max\{|r| - a, 0\}.$$

1.1.3 Log Barrier

$$\phi(u) = \begin{cases} -\log(1 - u^2), & |u| < 1 \\ \infty, & |u| \geq 1 \end{cases}$$

1.1.4 Huber Penalty

2 Moreau Envelope

Definition 2.1 (Moreau Envelope): Given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we have the Moreau Envelope as

$$f_u(\mathbf{x}) = \inf_{\mathbf{y} \in \mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u} \|\mathbf{x} - \mathbf{y}\|^2 \right\}.$$

Remark 2.2: The above expression is essentially an infimal convolution of $uf : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ where $g = (1/2)\|\cdot\|_2^2$.

Definition 2.3 (Proximal Operator): The proximal operator $\text{prox}_u(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of f with parameter u is defined as,

$$\text{prox}_u(\mathbf{x}) = \arg \min_{\mathbf{y} \in \mathbb{R}^n} \left\{ f(\mathbf{y}) + \frac{1}{2u} \|\mathbf{x} - \mathbf{y}\|^2 \right\}.$$

2.0.1 Properties of Moreau Envelope

1. $f_u(\mathbf{x})$ is differentiable if f is continuous. Further, the derivative is continuous.
2. $f_u(\mathbf{x})$ is defined for all $\mathbf{x} \in \mathbb{R}^n$, even when f is not.
3. $f_u(\mathbf{x})$ is convex whenever f is convex.
4. If x^* is the unique minimizer of f , then it is the unique minimizer of $f_u(\mathbf{x})$

Remark 2.4: $f_u(\mathbf{x})$ is convex, since it is the min over one variable of a jointly convex function.

Remark 2.5: $f_u(\mathbf{x})$ is continuously smooth.

Remark 2.6:

$$\text{epi} f_u(x) = \text{epi} f(x) \oplus \text{epi} \left(\frac{1}{2u} \|x\|^2 \right)$$

Remark 2.7: x^* is a minima of f if and only if x^* is a minima of f_u .

Remark 2.8:

$$\nabla f_u(\mathbf{x}) = \frac{1}{u}(x - \text{prox}_u(x)).$$

3 Epigraphs

TODO: complete this

Duality

Supporting Hyperplane Theorem

Definition 3.1 (Conjugate of a function): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. The conjugate $f^* : \mathbb{R}^n \rightarrow \mathbb{R}$ of f is defined as,

$$f^*(\mathbf{y}) = \sup_{\mathbf{x} \in \text{dom} f} (\mathbf{y}^T \mathbf{x} - f(\mathbf{x})).$$

Remark 3.2: The conjugate of f is also called Legendre transform, polar of f , Fenchel dual.

Remark 3.3: f^* is convex, irrespective of f .

Remark 3.4:

$$-f^*(0) = \min_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x})$$

Theorem 3.5 (Fenchel Young inequality):

$$f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T \mathbf{x}$$

3.1 Recovering f from f^*

Definition 3.6 (Biconjugate): Let $f^* : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be the conjugate of a function $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$. The **biconjugate** $f^{**} : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is defined as the conjugate of f^* ,

$$f^{**}(\mathbf{x}) = (f^*)^*(\mathbf{x}) = \sup_{\mathbf{y} \in \mathbb{R}^n} (\mathbf{y}^T \mathbf{x} - f^*(\mathbf{y})).$$

Remark 3.7 (Geometric Interpretation): The biconjugate f^{**} constructs the pointwise supremum (upper envelope) of all affine minorants of f . Specifically, the epigraph of f^{**} is the **closed convex hull** of the epigraph of f ,

$$\text{epi } f^{**} = \text{cl}(\text{conv}(\text{epi } f)).$$

Thus, f^{**} represents the "tightest" closed convex function that minorizes f .

3.1.1 Properties of the Biconjugate

Corollary 3.8 (Weak Duality): For any proper function f , the inequality $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ holds for all $\mathbf{x} \in \mathbb{R}^n$.

Proof. Recall the **Fenchel-Young inequality**, which states that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$f^*(\mathbf{y}) + f(\mathbf{x}) \geq \mathbf{x}^T \mathbf{y}.$$

Rearranging terms yields:

$$\mathbf{x}^T \mathbf{y} - f^*(\mathbf{y}) \leq f(\mathbf{x}).$$

Since this inequality holds for all $\mathbf{y} \in \mathbb{R}^n$, it must also hold for the supremum over $\mathbf{y}$:

$$\sup_{\mathbf{y} \in \mathbb{R}^n} (\mathbf{x}^T \mathbf{y} - f^*(\mathbf{y})) \leq f(\mathbf{x}).$$

By definition of the biconjugate, this is exactly $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$. □

Theorem 3.9 (Fenchel-Moreau): Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper function. Then $f = f^{**}$ if and only if f is **closed** (lower semi-continuous) and **convex**.

Proof. The forward implication is immediate: since f^{**} is defined as the pointwise supremum of a family of affine functions, it is necessarily closed and convex. Thus, if $f = f^{**}$, f must share these properties.

For the reverse implication, assume f is proper, closed, and convex. By the weak duality lemma, $f^{**}(\mathbf{x}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$. It remains to show that $f(\mathbf{x}) \leq f^{**}(\mathbf{x})$.

We proceed by contradiction. Suppose there exists a point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$. This implies that the point $(\mathbf{x}_0, f^{**}(\mathbf{x}_0)) \in \mathbb{R}^{n+1}$ lies strictly below the epigraph of f , denoted $\text{epi}(f)$. Since f is closed and convex, $\text{epi}(f)$ is a closed convex set.

By the **Strict Separating Hyperplane Theorem**, there exists a non-zero vector $(\mathbf{y}, \mu) \in \mathbb{R}^n \times \mathbb{R}$ and a scalar $\alpha \in \mathbb{R}$ such that:

$$\mathbf{y}^T \mathbf{x}_0 + \mu f^{**}(\mathbf{x}_0) > \alpha > \mathbf{y}^T \mathbf{x} + \mu t, \quad \forall (\mathbf{x}, t) \in \text{epi}(f).$$

Since $(\mathbf{x}, t) \in \text{epi}(f)$ allows t to be arbitrarily large, we must have $\mu \leq 0$ (otherwise the RHS would be unbounded).

Case 1: Non-vertical separation ($\mu < 0$) Divide the inequality by $-\mu > 0$. Let $\mathbf{m} = \mathbf{y}/(-\mu)$ and $c = \alpha/(-\mu)$. The separation becomes:

$$\mathbf{m}^T \mathbf{x}_0 - f^{**}(\mathbf{x}_0) > c > \mathbf{m}^T \mathbf{x} - t, \quad \forall (\mathbf{x}, t) \in \text{epi}(f).$$

Setting $t = f(\mathbf{x})$ for $\mathbf{x} \in \text{dom}(f)$, we obtain $\mathbf{m}^T \mathbf{x} - f(\mathbf{x}) < c$. Taking the supremum over $\mathbf{x}$:

$$f^*(\mathbf{m}) = \sup_{\mathbf{x}} (\mathbf{m}^T \mathbf{x} - f(\mathbf{x})) \leq c.$$

However, the left side of the separation inequality yields $\mathbf{m}^T \mathbf{x}_0 - c > f^{**}(\mathbf{x}_0)$. This implies that the affine function $h(\mathbf{z}) = \mathbf{m}^T \mathbf{z} - f^*(\mathbf{m})$ satisfies:

$$h(\mathbf{x}_0) = \mathbf{m}^T \mathbf{x}_0 - f^*(\mathbf{m}) \geq \mathbf{m}^T \mathbf{x}_0 - c > f^{**}(\mathbf{x}_0).$$

This contradicts the definition of f^{**} as the supremum of *all* such affine minorants: $f^{**}(\mathbf{x}_0) \geq h(\mathbf{x}_0)$.

Case 2: Vertical separation ($\mu = 0$) If $\mu = 0$, the separation reduces to $\mathbf{y}^T \mathbf{x}_0 > \alpha > \mathbf{y}^T \mathbf{x}$ for all $\mathbf{x} \in \text{dom}(f)$. This means $\mathbf{x}_0$ is strictly separated from the domain of f in $\mathbb{R}^n$. Since f is proper and convex, it admits at least one affine minorant $g(\mathbf{z}) = \mathbf{a}^T \mathbf{z} - b \leq f(\mathbf{z})$. Consider the function $g_k(\mathbf{z}) = g(\mathbf{z}) + k(\mathbf{y}^T \mathbf{z} - \alpha)$. For $\mathbf{x} \in \text{dom}(f)$, since $\mathbf{y}^T \mathbf{x} < \alpha$, the term $k(\mathbf{y}^T \mathbf{x} - \alpha)$ is negative. Thus, $g_k(\mathbf{x}) \leq g(\mathbf{x}) \leq f(\mathbf{x})$ for all $k > 0$. This means g_k is an affine minorant of f for any k . At $\mathbf{x}_0$, however, $\mathbf{y}^T \mathbf{x}_0 > \alpha$, so the term $k(\mathbf{y}^T \mathbf{x}_0 - \alpha) \rightarrow +\infty$ as $k \rightarrow \infty$. Consequently, $f^{**}(\mathbf{x}_0) \geq g_k(\mathbf{x}_0) \rightarrow +\infty$. Since we assumed $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0) < \infty$ (if $f(\mathbf{x}_0) = \infty$ the inequality holds trivially), this yields a contradiction.

Thus, the assumption $f^{**}(\mathbf{x}_0) < f(\mathbf{x}_0)$ is false. We conclude $f = f^{**}$. $\square$

Remark 3.10: If f is convex but not closed, f^{**} is the lower semi-continuous closure of f . If f is not convex, f^{**} is the convex envelope of f .

4 Duality

Corollary 4.1 (Fenchel-Young-Corollary): Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex and differentiable function. For any $\mathbf{x}, \mathbf{m} \in \mathbb{R}^n$, the following statements are equivalent:

1. $\mathbf{m} = \nabla f(\mathbf{x})$
2. $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T \mathbf{x}$

Proof. Recall the Fenchel-Young inequality, which states $f(\mathbf{x}) + f^*(\mathbf{m}) \geq \mathbf{m}^T \mathbf{x}$ for all $\mathbf{x}, \mathbf{m}$. We prove the equivalence of the equality condition and the gradient condition.

(1 $\implies$ 2): Assume $\mathbf{m} = \nabla f(\mathbf{x})$. Since f is convex and differentiable, the first-order condition for convexity states that for all $\mathbf{y} \in \mathbb{R}^n$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}).$$

Substituting $\mathbf{m} = \nabla f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T \mathbf{y} - \mathbf{m}^T \mathbf{x}.$$

Rearranging terms to isolate $\mathbf{m}^T \mathbf{y} - f(\mathbf{y})$,

$$\mathbf{m}^T \mathbf{y} - f(\mathbf{y}) \leq \mathbf{m}^T \mathbf{x} - f(\mathbf{x}).$$

Taking the supremum of the left-hand side over all $\mathbf{y} \in \mathbb{R}^n$ yields the definition of the conjugate $f^*(\mathbf{m})$,

$$f^*(\mathbf{m}) = \sup_{\mathbf{y}} (\mathbf{m}^T \mathbf{y} - f(\mathbf{y})) \leq \mathbf{m}^T \mathbf{x} - f(\mathbf{x}).$$

This implies $f(\mathbf{x}) + f^*(\mathbf{m}) \leq \mathbf{m}^T \mathbf{x}$. Combining this with the Fenchel-Young inequality (which gives the reverse inequality $\geq$), we obtain equality,

$$f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T \mathbf{x}.$$

(2 $\implies$ 1): Assume $f(\mathbf{x}) + f^*(\mathbf{m}) = \mathbf{m}^T \mathbf{x}$. Rearranging for $f(\mathbf{x})$,

$$f(\mathbf{x}) = \mathbf{m}^T \mathbf{x} - f^*(\mathbf{m}).$$

By definition of the conjugate, $f^*(\mathbf{m}) \geq \mathbf{m}^T \mathbf{y} - f(\mathbf{y})$ for any $\mathbf{y} \in \mathbb{R}^n$. Substituting this inequality into the equation above,

$$f(\mathbf{x}) \leq \mathbf{m}^T \mathbf{x} - (\mathbf{m}^T \mathbf{y} - f(\mathbf{y})).$$

Rearranging terms to compare $f(\mathbf{y})$ and $f(\mathbf{x})$,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \mathbf{m}^T (\mathbf{y} - \mathbf{x}).$$

This inequality holds for all $\mathbf{y} \in \mathbb{R}^n$. This is precisely the defining inequality for $\mathbf{m}$ being a subgradient of f at $\mathbf{x}$. Since f is differentiable, the subgradient is unique and equal to the gradient. Thus,

$$\mathbf{m} = \nabla f(\mathbf{x}).$$

□

Remark 4.2: Geometrically, this corollary asserts that the vector $\mathbf{m}$ defines a supporting hyperplane to the graph of f at $(\mathbf{x}, f(\mathbf{x}))$ if and only if Fenchel's inequality holds with equality.

4.1 Partial Minimization & Relation To Conjugate

Let $f : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R} \cup \{+\infty\}$ be a function of two variables $(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^k$.

Definition 4.3 (Partial Minimization): The partial minimization of f with respect to $\mathbf{x}$ is the function $h : \mathbb{R}^k \rightarrow \mathbb{R} \cup \{+\infty\}$ defined as,

$$h(\mathbf{y}) = \inf_{\mathbf{x} \in \mathbb{R}^n} f(\mathbf{x}, \mathbf{y}).$$

Lemma 4.4 (Preservation of Convexity): If f is jointly convex in $(\mathbf{x}, \mathbf{y})$, then $h(\mathbf{y})$ is a convex function.

Remark 4.5:

$$f^*(\mathbf{m}_1, \mathbf{m}_2) = \sup_{\mathbf{x}, \mathbf{y}} (\mathbf{m}_1^T \mathbf{x} + \mathbf{m}_2^T \mathbf{y} - f(\mathbf{x}, \mathbf{y}))$$

Remark 4.6: $h^*(\mathbf{m}) = f^*(0, \mathbf{m})$

Proof.

$$\begin{aligned} h^*(\mathbf{m}) &= \sup_{\mathbf{y}} (\mathbf{m}^T \mathbf{y} - h(\mathbf{y})) \\ &= \sup_{\mathbf{y}} \left(\mathbf{m}^T \mathbf{y} - \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) \right) \\ &= \sup_{\mathbf{y}} \sup_{\mathbf{x}} (\mathbf{m}^T \mathbf{y} - f(\mathbf{x}, \mathbf{y})) = f^*(0, \mathbf{m}) \end{aligned}$$

□

Remark 4.7: $\sup [-f^*(0, \mathbf{m})] \leq \inf f(\mathbf{x}, 0)$ i.e. $h^{**}(0) \leq h(0)$

Remark 4.8: Partial minimization appears naturally when eliminating primal variables in dual constructions.

4.1.1 Perturbation Interpretation Of Duality

For any problem $p^* = \min_{\mathbf{x}} f(\mathbf{x})$, "the" dual is constructed by

1. Introduce perturbations $\mathbf{y}$, and extend f such that $f(\mathbf{x}, 0)$ returns the function to minimize
2. the dual is $d^* = \sup [-f^*(0, \mathbf{m})]$

where $\mathbf{m}$ is the dual variable and $\mathbf{x}$ is the primal variable.

We have

$$d^* \leq p^*,$$

under convexity + mild conditions, we have strong duality, i.e. $d^* = p^*$.

5 Lagrangian Duality

Given the optimization problem in standard form:

$$p^* = \inf \{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, x \in \mathcal{D}\}$$

where $\mathcal{D} \subseteq \mathbb{R}^n$ is the intersection of the domains of all functions f_i .

5.1 The Lagrangian and Dual Function

Define the **Lagrangian** $L : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ for $\lambda \succeq 0$ as,

$$L(x, \lambda) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x).$$

The **Lagrange dual function** $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is defined as the pointwise infimum of the Lagrangian over x ,

$$g(\lambda) = \inf_{x \in \mathcal{D}} L(x, \lambda).$$

By the property of pointwise infima, g is a concave function, and it provides a lower bound on the primal optimal value: $g(\lambda) \leq p^*$ for all $\lambda \succeq 0$.

The Dual Problem

The **Lagrange dual problem** seeks the tightest lower bound,

$$\begin{aligned} d^* = \sup_{\lambda} \quad & g(\lambda) \\ \text{s.t.} \quad & \lambda \succeq 0 \end{aligned}$$

The relationship between the primal and dual optimal values is governed by **Weak Duality** ($d^* \leq p^*$). Under constraint qualifications (e.g., Slater's condition) and convexity, **Strong Duality** holds ($d^* = p^*$).

6 Shadow Price Interpretation of Duality

7 Game Theoretic Interpretation of Duality

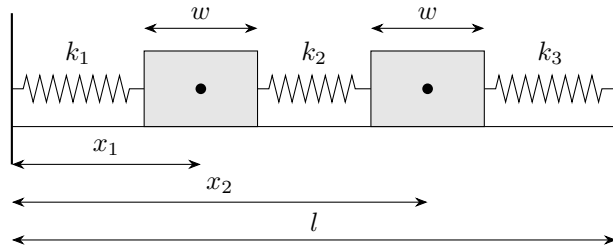
8 KKT Conditions

Consider a standard optimization problem with strong duality and differentiable constraints. Let x^* be the primal optima and λ^* be the dual optima.

Definition 8.1 (KKT Conditions): *The following statements constitute the KKT conditions,*

1. $f_i(x^*) \leq 0$ & $h_j(x^*) = 0, \forall i, j$ (Primal feasibility).
2. $\lambda_i^* \geq 0, \forall i$ (Dual feasibility).
3. $\lambda_i^* f_i(x^*) = 0, \forall i$ (Complementary slackness).
4. $\nabla_x L(x^*, \lambda) = 0$.

9 Mechanistic Interpretation of KKT Conditions



There's maybe no need for this.

Yeah, there's no need for this.

10 Multicriterion Optimization

Some shit. Write it.

11 Moreau Envelope (Continued)

Continued from [here](#).

Example

$$p^* = \min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2 \quad g(x) = p^*$$

Reframe as:

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}} \quad & \|\mathbf{y}\|_2 \\ \text{s.t.} \quad & \mathbf{Ax} - \mathbf{b} = \mathbf{y} \end{aligned}$$

Lagrangian:

$$L(\mathbf{x}, \mathbf{y}, \mathbf{v}) = \|\mathbf{y}\|_2 + \mathbf{v}^T (\mathbf{Ax} - \mathbf{b} - \mathbf{y})$$

Dual Function:

$$\begin{aligned} g(\mathbf{v}) &= \inf_{\mathbf{x}, \mathbf{y}} [\|\mathbf{y}\|_2 + \mathbf{v}^T (\mathbf{Ax} - \mathbf{b} - \mathbf{y})] \\ &= -\mathbf{v}^T \mathbf{b} + \left[\inf_{\mathbf{x}, \mathbf{y}} \left(\underbrace{\|\mathbf{y}\|_2 - \mathbf{v}^T \mathbf{y}}_{\substack{-\infty \text{ if } \|\mathbf{v}\|_2 > 1 \\ 0 \text{ otherwise}}} \right) + \left(\underbrace{(\mathbf{A}^T \mathbf{v})^T \mathbf{x}}_{\substack{0 \text{ if } \mathbf{A}^T \mathbf{v} = 0 \\ -\infty \text{ otherwise}}} \right) \right] \end{aligned}$$

Result:

$$g(\mathbf{v}) = \begin{cases} -\mathbf{v}^T \mathbf{b} & \text{if } \mathbf{A}^T \mathbf{v} = 0 \text{ \& } \|\mathbf{v}\|_2 \leq 1 \\ -\infty & \text{otherwise} \end{cases}$$

11.1 Dual of Moreau envelope

Let $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, closed, convex function. Recall the Moreau envelope $f_u(x)$ with parameter $u > 0$ is defined as the infimal convolution of f and a quadratic kernel,

$$f_u(x) = \inf_y \left\{ f(y) + \frac{1}{2u} \|x - y\|^2 \right\}.$$

To derive its dual representation, we rewrite this as a constrained minimization problem by introducing an auxiliary variable z

$$\begin{aligned} \min_{y, z} \quad & f(y) + \frac{1}{2u} \|z\|^2 \\ \text{s.t.} \quad & x - y = z. \end{aligned}$$

The dual function $g(v)$ is given by

$$g(v) = \langle v, x \rangle - f^*(v) - \frac{u}{2} \|v\|^2,$$

which leads to the identity

$$f_u(x) = \left(f^* + \frac{u}{2} \|\cdot\|^2 \right)^*(x).$$

Proof. We form the Lagrangian $L(y, z, v)$ with dual variable $v \in \mathbb{R}^n$,

$$L(y, z, v) = f(y) + \frac{1}{2u} \|z\|^2 + \langle v, x - y - z \rangle.$$

The Lagrange dual function $g(v)$ is obtained by minimizing the Lagrangian with respect to the primal variables y and z ,

$$\begin{aligned} g(v) &= \inf_{y, z} L(y, z, v) \\ &= \inf_{y, z} \left\{ f(y) + \frac{1}{2u} \|z\|^2 + \langle v, x \rangle - \langle v, y \rangle - \langle v, z \rangle \right\} \\ &= \langle v, x \rangle + \inf_y \{ f(y) - \langle v, y \rangle \} + \inf_z \left\{ \frac{1}{2u} \|z\|^2 - \langle v, z \rangle \right\}. \end{aligned}$$

We evaluate the infima term-wise:

1. Term involving y : Using the definition of the Fenchel conjugate, $f^*(v) = \sup_y \{ \langle v, y \rangle - f(y) \}$, we have,

$$\inf_y \{ f(y) - \langle v, y \rangle \} = -\sup_y \{ \langle v, y \rangle - f(y) \} = -f^*(v).$$

2. Term involving z : This is the conjugate of the quadratic function $h(z) = \frac{1}{2u} \|z\|^2$. Setting the gradient with respect to z to zero,

$$\nabla_z \left(\frac{1}{2u} \|z\|^2 - \langle v, z \rangle \right) = \frac{z}{u} - v = 0 \implies z = uv.$$

Substituting $z = uv$ back into the expression,

$$\frac{1}{2u}\|uv\|^2 - \langle v, uv \rangle = \frac{u}{2}\|v\|^2 - u\|v\|^2 = -\frac{u}{2}\|v\|^2.$$

Combining these, the dual function

$$g(v) = \langle v, x \rangle - f^*(v) - \frac{u}{2}\|v\|^2.$$

The dual problem consists of maximizing $g(v)$. Since f is proper, closed, and convex, strong duality holds ($p^* = d^*$), and we can express the primal optimum as the supremum of the dual:

$$\begin{aligned} f_u(x) &= \sup_v g(v) \\ &= \sup_v \left\{ \langle v, x \rangle - \left(f^*(v) + \frac{u}{2}\|v\|^2 \right) \right\}. \end{aligned}$$

Recognizing the right-hand side as the Fenchel conjugate of the function $\phi(v) = f^*(v) + \frac{u}{2}\|v\|^2$ evaluated at x , we conclude:

$$f_u(x) = \left(f^* + \frac{u}{2}\|\cdot\|^2 \right)^*(x).$$

□

11.2 Moreau Decomposition

12 Proximal Algorithms

12.1 Fixed Point

13 Tutorial 1: Projection on ℓ_1 ball

14 Statistical Interpretation of Norm Approximation

15 Analysis of Least Squares

15.1 Problem Setup and Definitions

Let us define a dataset comprising a fixed design matrix $\mathbf{A} \in \mathbb{R}^{n \times d}$ (where $n > d$) representing the feature vectors, and a random response vector $\mathbf{y} \in \mathbb{R}^n$. We assume the labels are generated according to a linear ground-truth model:

$$\mathbf{y} = \mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon} \tag{1}$$

where $\mathbf{x}^* \in \mathbb{R}^d$ is the true, unknown parameter vector, and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ is the observation noise vector. The goal is to learn an estimator $\hat{\mathbf{x}}$ that closely approximates $\mathbf{y} \approx \mathbf{A}\hat{\mathbf{x}}$.

Definition 15.1 (Risk and Excess Risk): Let $\tilde{\mathbf{y}} = \mathbf{A}\mathbf{x}^* + \tilde{\epsilon}$ be an independent test sample drawn from the identical data-generating process, where $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ is independent of ϵ . The **Risk** of an estimator $\hat{\mathbf{x}}$ is defined as:

$$\mathcal{R}(\hat{\mathbf{x}}) = \frac{1}{n} \mathbb{E}_{\epsilon, \tilde{\epsilon}} \|\tilde{\mathbf{y}} - \mathbf{A}\hat{\mathbf{x}}\|^2 \quad (2)$$

The **Excess Risk** isolates the error attributable to the estimation of $\mathbf{x}^*$ by removing the irreducible noise σ^2 :

$$\mathcal{E}(\hat{\mathbf{x}}) = \frac{1}{n} \mathbb{E}_{\epsilon} \|\mathbf{A}(\mathbf{x}^* - \hat{\mathbf{x}})\|^2 \quad (3)$$

Lemma 15.2 (Bias-Variance Decomposition): The excess risk of any estimator $\hat{\mathbf{x}}$ can be additively decomposed into an approximation error (Bias term) and a prediction error (Variance term):

$$\mathcal{E}(\hat{\mathbf{x}}) = \underbrace{\frac{1}{n} \mathbb{E}_{\epsilon} \|\mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}])\|^2}_{\text{Bias}} + \underbrace{\frac{1}{n} \mathbb{E}_{\epsilon} \|\mathbf{A}(\hat{\mathbf{x}} - \mathbb{E}[\hat{\mathbf{x}}])\|^2}_{\text{Variance}} \quad (4)$$

Proof. By adding and subtracting $\mathbb{E}[\hat{\mathbf{x}}]$, we have $\mathbf{A}(\mathbf{x}^* - \hat{\mathbf{x}}) = \mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}]) + \mathbf{A}(\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}})$. Taking the expected squared norm, the cross-term $\frac{2}{n} \mathbb{E}[(\mathbf{A}(\mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}]))^\top \mathbf{A}(\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}})]$ vanishes because $\mathbb{E}[\mathbb{E}[\hat{\mathbf{x}}] - \hat{\mathbf{x}}] = \mathbf{0}$. The remaining terms form the bias and variance components. $\square$

15.2 Analysis of Ordinary Least Squares (OLS)

Theorem 15.3 (Risk of OLS): The Ordinary Least Squares estimator, defined as $\hat{\mathbf{x}}_{\text{OLS}} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2$, is an unbiased estimator with an excess risk of exactly $\frac{\sigma^2 d}{n}$.

Proof. The closed-form solution is $\hat{\mathbf{x}}_{\text{OLS}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y}$. Taking the expectation:

$$\mathbb{E}[\hat{\mathbf{x}}_{\text{OLS}}] = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbb{E}[\mathbf{A}\mathbf{x}^* + \epsilon] = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{A}\mathbf{x}^* = \mathbf{x}^*$$

Since the estimator is unbiased, the excess risk is entirely variance. As derived in standard regression analysis, the variance evaluates to $\frac{\sigma^2 d}{n}$. $\square$

15.3 Analysis of Ridge Regression

To mitigate the variance term characteristic of OLS, particularly in settings where d is large relative to n , we introduce ℓ_2 regularization.

Definition 15.4 (Ridge Estimator): For a regularization parameter $\lambda > 0$, the Ridge estimator is:

$$\hat{\mathbf{x}}_{\text{ridge}} = \arg \min_{\mathbf{x}} \left[\frac{1}{n} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2 + \lambda \|\mathbf{x}\|^2 \right] = \frac{1}{n} (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y} \quad (5)$$

where $\mathbf{\Sigma} = \frac{1}{n} \mathbf{A}^\top \mathbf{A}$ is the empirical covariance matrix.

By substituting $\mathbf{y} = \mathbf{A}\mathbf{x}^* + \epsilon$, the expected Ridge estimator is $\mathbb{E}[\hat{\mathbf{x}}_{\text{ridge}}] = (\mathbf{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{\Sigma} \mathbf{x}^*$, demonstrating that Ridge regression introduces bias.

Lemma 15.5 (Ridge Bias Bound): The bias term for the Ridge estimator is bounded by $\frac{\lambda}{4} \|\mathbf{x}^*\|^2$.

Proof. Let

$$\Delta \mathbf{x} = \mathbf{x}^* - \mathbb{E}[\hat{\mathbf{x}}_{\text{ridge}}] = (\mathbf{I} - (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \boldsymbol{\Sigma}) \mathbf{x}^* = \lambda (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{x}^*.$$

The bias is:

$$\text{Bias} = \frac{1}{n} \|\mathbf{A} \Delta \mathbf{x}\|^2 = (\Delta \mathbf{x})^\top \boldsymbol{\Sigma} (\Delta \mathbf{x}) = \lambda^2 (\mathbf{x}^*)^\top (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \boldsymbol{\Sigma} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{x}^*.$$

Because $\boldsymbol{\Sigma}$ and $(\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1}$ commute, we rewrite this as $(\mathbf{x}^*)^\top [\lambda^2 \boldsymbol{\Sigma} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-2}] \mathbf{x}^*$. Let $\boldsymbol{\Sigma} = \mathbf{V} \mathbf{D} \mathbf{V}^\top$ be the eigendecomposition with eigenvalues $\mu_i \geq 0$. The eigenvalues of the bracketed matrix are $f(\mu_i) = \frac{\lambda^2 \mu_i}{(\mu_i + \lambda)^2}$. Using the identity $(\mu_i + \lambda)^2 = (\mu_i - \lambda)^2 + 4\mu_i \lambda \geq 4\mu_i \lambda$, we have $f(\mu_i) \leq \frac{\lambda^2 \mu_i}{4\mu_i \lambda} = \frac{\lambda}{4}$. By the Rayleigh quotient, the quadratic form is bounded by the maximum eigenvalue times $\|\mathbf{x}^*\|^2$:

$$\text{Bias} \leq \frac{\lambda}{4} \|\mathbf{x}^*\|^2$$

□

Lemma 15.6 (Ridge Variance Bound): *The variance term for the Ridge estimator is bounded by $\frac{\sigma^2 \text{Tr}(\boldsymbol{\Sigma})}{4n\lambda}$.*

Proof. The deviation is $\hat{\mathbf{x}}_{\text{ridge}} - \mathbb{E}[\hat{\mathbf{x}}_{\text{ridge}}] = \frac{1}{n} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \boldsymbol{\epsilon}$. The variance is $\mathbb{E}[\|\mathbf{C} \boldsymbol{\epsilon}\|^2]$ where $\mathbf{C} = \frac{1}{n\sqrt{n}} \mathbf{A} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top$. For any fixed matrix $\mathbf{C}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, $\mathbb{E} \|\mathbf{C} \boldsymbol{\epsilon}\|^2 = \sigma^2 \text{Tr}(\mathbf{C}^\top \mathbf{C})$. Evaluating the trace:

$$\begin{aligned} \text{Variance} &= \sigma^2 \text{Tr} \left(\frac{1}{n^3} \mathbf{A} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{A} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \right) \\ &= \frac{\sigma^2}{n} \text{Tr} \left(\frac{\mathbf{A}^\top \mathbf{A}}{n} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \frac{\mathbf{A}^\top \mathbf{A}}{n} (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} \right) = \frac{\sigma^2}{n} \text{Tr} (\boldsymbol{\Sigma}^2 (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-2}) \end{aligned}$$

The trace is the sum of eigenvalues $\sum_i \frac{\mu_i^2}{(\mu_i + \lambda)^2}$. Since $\frac{\mu_i}{(\mu_i + \lambda)^2} \leq \frac{1}{4\lambda}$ as shown previously:

$$\sum_i \mu_i \left(\frac{\mu_i}{(\mu_i + \lambda)^2} \right) \leq \sum_i \mu_i \left(\frac{1}{4\lambda} \right) = \frac{\text{Tr}(\boldsymbol{\Sigma})}{4\lambda}$$

Thus, $\text{Variance} \leq \frac{\sigma^2 \text{Tr}(\boldsymbol{\Sigma})}{4n\lambda}$. □

Theorem 15.7 (Excess Risk of Ridge Regression): *By selecting the optimal regularization parameter $\lambda = \frac{\sigma \sqrt{\text{Tr}(\boldsymbol{\Sigma})}}{\|\mathbf{x}^*\| \sqrt{n}}$, the excess risk of the Ridge estimator is bounded by $\mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$.*

Proof. Combining the lemmas, the upper bound on the excess risk is:

$$\mathcal{E}(\hat{\mathbf{x}}_{\text{ridge}}) \leq \frac{\|\mathbf{x}^*\|^2}{4} \lambda + \frac{\sigma^2 \text{Tr}(\boldsymbol{\Sigma})}{4n} \frac{1}{\lambda}$$

This bound is of the form $f(\lambda) = a\lambda + \frac{b}{\lambda}$. By the AM-GM inequality ($a\lambda + \frac{b}{\lambda} \geq 2\sqrt{ab}$), the minimum is achieved at $\lambda = \sqrt{b/a}$. Substituting $a = \frac{\|\mathbf{x}^*\|^2}{4}$ and $b = \frac{\sigma^2 \text{Tr}(\mathbf{\Sigma})}{4n}$, we find the optimal λ . The corresponding minimal risk bound is $2\sqrt{ab}$, which evaluates to:

$$\min_{\lambda > 0} \mathcal{E}(\hat{\mathbf{x}}_{\text{ridge}}) \leq \frac{\sigma \|\mathbf{x}^*\| \sqrt{\text{Tr}(\mathbf{\Sigma})}}{2\sqrt{n}}$$

□

16 Gradient Descent

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a continuously differentiable objective function. We consider the unconstrained optimization problem $\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$. The standard Gradient Descent (GD) algorithm generates a sequence of iterates $\{\mathbf{x}^{(k)}\}_{k=0}^{\infty}$ via the recurrence relation

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta \nabla f(\mathbf{x}^{(k)}) \quad (6)$$

where $\eta > 0$ is the step size. To rigorously guarantee convergence, we must impose structural conditions on the objective function f .

Definition 16.1 (*L-Smoothness and Strong Convexity*): A differentiable function f is *L-smooth* if its gradient is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d \quad (7)$$

Furthermore, f is *μ -strongly convex* for $\mu > 0$ if:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|_2^2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d \quad (8)$$

If f is twice continuously differentiable, these conditions are equivalent to bounding the spectrum of the Hessian matrix: $\mu \mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L \mathbf{I}$ for all $\mathbf{x} \in \mathbb{R}^d$, where $\preceq$ denotes the Loewner partial order.

16.1 The Least Squares Objective

Let $\mathbf{A} \in \mathbb{R}^{n \times d}$ ($n \geq d$) denote a fixed design matrix and $\mathbf{y} \in \mathbb{R}^n$ a target vector. The normalized ordinary least squares (OLS) problem seeks to minimize the empirical risk:

$$f(\mathbf{x}) = \frac{1}{2n} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 \quad (9)$$

Lemma 16.2 (*Properties of the OLS Objective*): Assume $\mathbf{A}$ has full column rank ($\text{rank}(\mathbf{A}) = d$). Let $\mathbf{H} \triangleq \frac{1}{n} \mathbf{A}^\top \mathbf{A}$ be the Hessian matrix. The objective $f(\mathbf{x})$ is *L-smooth* and *μ -strongly convex* with parameters:

$$L = \lambda_{\max}(\mathbf{H}), \quad \mu = \lambda_{\min}(\mathbf{H}) \quad (10)$$

where $\lambda_{\max}$ and $\lambda_{\min}$ denote the maximal and minimal eigenvalues of $\mathbf{H}$, respectively. Moreover, the unique global minimizer is $\mathbf{x}^* = \mathbf{H}^{-1} \left(\frac{1}{n} \mathbf{A}^\top \mathbf{y} \right)$.

Proof. The gradient is explicitly given by $\nabla f(\mathbf{x}) = \mathbf{H}\mathbf{x} - \frac{1}{n}\mathbf{A}^\top \mathbf{y}$. The Hessian is the constant symmetric matrix $\nabla^2 f(\mathbf{x}) = \mathbf{H}$. Since $\mathbf{A}$ has full column rank, $\mathbf{A}^\top \mathbf{A}$ is positive definite, implying $\mu = \lambda_{\min}(\mathbf{H}) > 0$. Thus, $\mu\mathbf{I} \preceq \mathbf{H} \preceq L\mathbf{I}$, fulfilling Definition **L-Smoothness and Strong Convexity** globally. Setting $\nabla f(\mathbf{x}^*) = \mathbf{0}$ directly yields the unique optimum due to the invertibility of $\mathbf{H}$. $\square$

16.2 Spectral Error Dynamics

We analyze the evolution of the error vector $\mathbf{e}^{(k)} \triangleq \mathbf{x}^{(k)} - \mathbf{x}^*$.

Theorem 16.3 (Exact Error Operator): *For the OLS objective, the error sequence under Gradient Descent evolves according to the linear operator $(\mathbf{I} - \eta\mathbf{H})$:*

$$\mathbf{e}^{(k)} = (\mathbf{I} - \eta\mathbf{H})^k \mathbf{e}^{(0)} \quad (11)$$

Proof. By substituting the gradient into the GD update rule:

$$\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \eta \left(\mathbf{H}\mathbf{x}^{(k-1)} - \frac{1}{n}\mathbf{A}^\top \mathbf{y} \right)$$

Utilizing the optimality condition $\frac{1}{n}\mathbf{A}^\top \mathbf{y} = \mathbf{H}\mathbf{x}^*$, we have:

$$\begin{aligned} \mathbf{x}^{(k)} &= \mathbf{x}^{(k-1)} - \eta\mathbf{H}(\mathbf{x}^{(k-1)} - \mathbf{x}^*) \\ \mathbf{x}^{(k)} - \mathbf{x}^* &= (\mathbf{I} - \eta\mathbf{H})(\mathbf{x}^{(k-1)} - \mathbf{x}^*) \end{aligned}$$

Applying this relation recursively k times yields the stated theorem. $\square$

To bound $\|\mathbf{e}^{(k)}\|_2$, we must constrain the spectral radius $\rho(\mathbf{I} - \eta\mathbf{H})$. By the Spectral Theorem, the symmetric positive definite matrix $\mathbf{H}$ admits an orthogonal eigendecomposition $\mathbf{H} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, where $\mathbf{V}$ is an orthogonal matrix and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d)$ contains the eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d > 0$.

Corollary 16.4 (Spectral Bound and Convergence): *The induced ℓ_2 -norm of the iteration operator is bounded by:*

$$\|\mathbf{I} - \eta\mathbf{H}\|_2 = \max_{1 \leq i \leq d} |1 - \eta\lambda_i| \quad (12)$$

Consequently, $\lim_{k \rightarrow \infty} \mathbf{x}^{(k)} = \mathbf{x}^*$ for any initialization $\mathbf{x}^{(0)}$ if and only if $0 < \eta < \frac{2}{L}$.

16.3 Optimal Step Size via Minimax Formulation

We define the condition number of the problem as $\kappa \triangleq \frac{L}{\mu} = \frac{\lambda_1}{\lambda_d} \geq 1$. The contraction rate is defined as $q(\eta) \triangleq \max_i |1 - \eta\lambda_i|$. To maximize the convergence speed, we seek the step size η^* that minimizes this worst-case spectral radius.

Theorem 16.5 (Optimal Constant Step Size): *The optimal fixed step size minimizing the spectral radius is:*

$$\eta^* = \frac{2}{L + \mu} \quad (13)$$

yielding an optimal contraction factor of $q(\eta^*) = \frac{\kappa-1}{\kappa+1}$.

Proof. We formulate the minimax optimization problem:

$$\eta^* = \arg \min_{\eta > 0} \max_{1 \leq i \leq d} |1 - \eta \lambda_i| \quad (14)$$

Because $1 - \eta \lambda$ is a strictly decreasing function of λ for any fixed $\eta > 0$, the maximum absolute value must occur at the boundaries of the spectrum: $\lambda_1 = L$ or $\lambda_d = \mu$. Thus:

$$\max_{1 \leq i \leq d} |1 - \eta \lambda_i| = \max(|1 - \eta \mu|, |1 - \eta L|) \quad (15)$$

For a valid learning rate ($\eta < 2/L$), the term $(1 - \eta \mu)$ is strictly positive and linearly decreasing, while $(1 - \eta L)$ becomes negative, making $|1 - \eta L| = \eta L - 1$ linearly increasing. The global minimum of the maximum of these two intersecting linear functions occurs precisely where they are equal:

$$1 - \eta \mu = \eta L - 1 \implies \eta(L + \mu) = 2 \implies \eta^* = \frac{2}{L + \mu}$$

Substituting η^* back into the contraction factor yields:

$$q(\eta^*) = 1 - \left(\frac{2}{L + \mu} \right) \mu = \frac{L - \mu}{L + \mu} = \frac{(L/\mu) - 1}{(L/\mu) + 1} = \frac{\kappa - 1}{\kappa + 1}$$

□

Remark 16.6 (Iteration Complexity): Given the optimal rate $q(\eta^*) = \frac{\kappa - 1}{\kappa + 1} \approx 1 - \frac{2}{\kappa}$ (for large κ), the error bound is $\|\mathbf{e}^{(k)}\|_2 \leq (1 - \frac{2}{\kappa})^k \|\mathbf{e}^{(0)}\|_2 \leq \exp(-\frac{2k}{\kappa}) \|\mathbf{e}^{(0)}\|_2$. To guarantee $\|\mathbf{e}^{(k)}\|_2 \leq \epsilon$, the required number of iterations scales as $\mathcal{O}(\kappa \log \frac{1}{\epsilon})$.

16.4 Computational Complexity: Exact Direct Methods

Finding the exact optimal solution $\mathbf{x}^*$ can be done via several matrix factorizations, all of which generally scale as $\mathcal{O}(nd^2)$ but differ in practical constants and numerical stability:

1. **Normal Equations (Cholesky):** $\mathbf{x}^* = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y}$. Computing the Hessian $\mathbf{A}^\top \mathbf{A}$ takes $\mathcal{O}(nd^2)$, and its inversion/factorization takes $\mathcal{O}(d^3)$. Total time is $\mathcal{O}(nd^2)$.
2. **QR Decomposition:** $\mathbf{A} = \mathbf{Q}\mathbf{R}$. The solution is obtained by solving the triangular system $\mathbf{R}\mathbf{x}^* = \mathbf{Q}^\top \mathbf{y}$, taking $\mathcal{O}(nd^2)$.
3. **Singular Value Decomposition (SVD):** $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$, implying $\mathbf{x}^* = \mathbf{V}\mathbf{\Sigma}^+ \mathbf{U}^\top \mathbf{y}$. This is the most numerically stable but computationally heaviest $\mathcal{O}(nd^2)$ approach.

In practice, execution times follow: Cholesky < QR < SVD.

16.5 Gradient Descent (GD) Iterations

GD is an approximate method requiring $k = \mathcal{O}(\kappa \log \frac{1}{\epsilon})$ iterations to reach ϵ -accuracy, where κ is the condition number of the Hessian $\mathbf{H} = \frac{1}{n} \mathbf{A}^\top \mathbf{A}$. We have two primary ways to compute the gradient step $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta \nabla f(\mathbf{x}^{(k)})$:

Lemma 16.7 (GD Complexity Regimes): *Gradient Descent can be implemented in two distinct computational paradigms:*

- **Approach 1 (Precomputation):** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta (\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top \mathbf{y})$. We precompute $\mathbf{H}$ and $\mathbf{A}^\top \mathbf{y}$ in $\mathcal{O}(nd^2)$. Each subsequent iteration takes $\mathcal{O}(d^2)$. Total complexity: $\mathcal{O}(nd^2 + d^2\kappa \log \frac{1}{\epsilon})$.
- **Approach 2 (No Precomputation):** $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \frac{\eta}{n}\mathbf{A}^\top (\mathbf{A}\mathbf{x}^{(k)} - \mathbf{y})$. We compute the matrix-vector products iteratively. Each iteration takes $\mathcal{O}(nd)$. Total complexity: $\mathcal{O}(nd\kappa \log \frac{1}{\epsilon})$.

Remark 16.8 (Why use Gradient Descent?): *In regimes where d is very large, Approach 2 avoids the $\mathcal{O}(nd^2)$ cost entirely, offering significant computational savings if the problem is well-conditioned (κ is small). Furthermore, GD is favored for:*

1. **Memory Restrictions:** *Iterative methods avoid storing the $d \times d$ dense Hessian, making them suitable for streaming algorithms or massive datasets.*
2. **Implicit Regularization:** *When stopped early, GD provides better generalization error by preventing overfitting to the noise, acting akin to Ridge regularization.*

16.6 Generalization Error and Early Stopping

We formalize the implicit regularization of GD by analyzing the noisy observation model. Let the training labels be generated by $\mathbf{y}_{\text{train}} = \mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

We obtain an estimate $\hat{\mathbf{x}}^{(k)}$ via k iterations of GD on $\mathbf{y}_{\text{train}}$. We wish to analyze the test error (Risk) on a new sample:

$$\mathcal{R}(k) = \frac{1}{n} \mathbb{E} \left\| \mathbf{A}\hat{\mathbf{x}}^{(k)} - \mathbf{y}_{\text{test}} \right\|_2^2$$

Recall that for the exact OLS estimator $\hat{\mathbf{x}}_{\text{OLS}}$, the risk is strictly $\mathcal{R}_{\text{OLS}} = \sigma^2 + \frac{\sigma^2 d}{n}$. We now track this risk strictly as a function of the iteration count k .

Theorem 16.9 (Noisy Error Dynamics): *Let $\mathbf{A}/\sqrt{n} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ be the thin SVD of the normalized design matrix, implying $\mathbf{H} = \mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^\top$. If we project the error into the eigenbasis $\boldsymbol{\alpha}^{(k)} = \mathbf{V}^\top \mathbf{x}^{(k)}$, the error recurrence with noise is:*

$$\boldsymbol{\alpha}^{(k+1)} - \boldsymbol{\alpha}^* = (\mathbf{I} - \eta \boldsymbol{\Sigma}^2)(\boldsymbol{\alpha}^{(k)} - \boldsymbol{\alpha}^*) + \frac{\eta}{\sqrt{n}} \boldsymbol{\Sigma} \tilde{\boldsymbol{\epsilon}} \quad (16)$$

where $\tilde{\boldsymbol{\epsilon}} = \mathbf{U}^\top \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ is the projected noise vector.

Proof. Substituting the noisy labels into the GD update:

$$\begin{aligned} \mathbf{x}^{(k+1)} &= \mathbf{x}^{(k)} - \eta \left(\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top \mathbf{y}_{\text{train}} \right) \\ &= \mathbf{x}^{(k)} - \eta \left(\mathbf{H}\mathbf{x}^{(k)} - \frac{1}{n}\mathbf{A}^\top (\mathbf{A}\mathbf{x}^* + \boldsymbol{\epsilon}) \right) \\ \mathbf{x}^{(k+1)} - \mathbf{x}^* &= (\mathbf{I} - \eta \mathbf{H})(\mathbf{x}^{(k)} - \mathbf{x}^*) + \frac{\eta}{n} \mathbf{A}^\top \boldsymbol{\epsilon} \end{aligned}$$

Using the thin SVD, the noise term maps to $\frac{\eta}{n}(\sqrt{n}\mathbf{V}\mathbf{\Sigma}\mathbf{U}^\top)\boldsymbol{\epsilon} = \frac{\eta}{\sqrt{n}}\mathbf{V}\mathbf{\Sigma}\tilde{\boldsymbol{\epsilon}}$. Multiplying the entire recurrence by $\mathbf{V}^\top$ to change into the $\boldsymbol{\alpha}$ basis yields:

$$\begin{aligned}\mathbf{V}^\top(\mathbf{x}^{(k+1)} - \mathbf{x}^*) &= \mathbf{V}^\top(\mathbf{I} - \eta\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top)(\mathbf{x}^{(k)} - \mathbf{x}^*) + \frac{\eta}{\sqrt{n}}\mathbf{\Sigma}\tilde{\boldsymbol{\epsilon}} \\ \boldsymbol{\alpha}^{(k+1)} - \boldsymbol{\alpha}^* &= (\mathbf{I} - \eta\mathbf{\Sigma}^2)(\boldsymbol{\alpha}^{(k)} - \boldsymbol{\alpha}^*) + \frac{\eta}{\sqrt{n}}\mathbf{\Sigma}\tilde{\boldsymbol{\epsilon}}\end{aligned}$$

Because $\mathbf{U}$ has orthonormal columns ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}_d$), the linear transformation of the isotropic Gaussian noise preserves its distribution: $\tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_d)$. $\square$